Doc. 314 – The Lattice in Hilbert Space: Geometry, Spectrum and Fibre

Johann Pascher

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Contents

Doc. 314 – The Lattice in Hilbert Space

What this is about

FFGFT computes geometrically on T^4 with the D4 lattice and quantum-mechanically on $\mathscr{H} = L^2(\mathrm{T}^4) \otimes \mathbb{C}^3$ (Docs. 230/282). The transition raises an obvious question: *do the lattices become flat?* Does anything remain of kissing number 24, packing density and triality, or is every lattice the same in Hilbert space?

The answer: the torus is flat anyway – curvature was never the difference – and the lattice structure is not lost but **migrates into the spectrum and into the fibre**. This document computes that through (three scripts, Ch. J) and finds:

1. **The spectrum separates what curvature does not [K]:** Z^4 starts at $|k|^2 = 1$ with degeneracy 8, D4 at $|k|^2 = 2$ with degeneracy 24; odd norms are entirely absent for D4. The first excitation carries exactly $24 = 12 + 12$ – FCC half and winding half separately addressable.
2. **Triality is built in, not attached [K]:** $|\mathrm{Aut}(D4)| = 1152 = |W(F_4)|$, the quotient to $W(D4)$ is $6 = |S_3|$; among the 80 order-3 elements, 16 generate exactly the T^4/Z_3 orbifold with $|\det(1 - A)| = 9$ fixed points and act on every mode orbit as a $\mathbb{C}^3$ circulant.
3. **A negative finding with theorem character [K]:** 5 does not divide $1152 = 2^7 \cdot 3^2$, so $\mathrm{Aut}(D4)$ contains *no* element of order 5. The fivefold rotation from which $\theta = 2/9$ arises (Doc. 293) is in principle incompatible with the lattice – the phase belongs to a different layer.
4. **The doubling is nevertheless prepared [K]:** $24 = 8 \times 3 = 4 \times 6$; four disjoint six-blocks with exactly the C_3 content that $3 \oplus 3'$ requires.

Not part of the A-series; status: working document. References: Docs. 230/282 ($\mathscr{H} = L^2(\mathrm{T}^4) \otimes \mathbb{C}^3$), 285 (doubling $6 = 3 \oplus 3'$), 291 (channel analysis), 293 ($\theta = 2/9$ icosahedral), 311 (Four onto Three, $24 = 12 + 12$).

Reader's guide: the thread of the argument

The question. In Hilbert space the lattice becomes the index set of the Fourier modes. It is then no longer a sphere packing – in that sense every lattice becomes "flat". The question is whether the information disappears with it.

The answer in one line. *Geometry → spectrum + fibre.* The degeneracies of the Laplacian are the theta series of the lattice; the kissing number becomes the degeneracy of the first excitation; triality becomes the orbifold and fibre structure.

What follows. First, two flat tori are distinguishable by their spectra (Ch. B). Second, the $\tilde{T} \cdot m = 1$ structure becomes spectrally readable when $R_4 \neq R_{123}$, and the level ordering changes at exactly two root radii (Ch. C). Third, the Z_3 need not be imposed – it is a lattice symmetry (Ch. E).

The boundary, drawn cleanly. Fivefold symmetry is impossible on the lattice (Ch. F). This does not hit the doubling, because that lives on the *fibre* and the interface of the two levels is exactly C_3 – occupied by triality (Ch. G). And the container structure holds on *every* shell, so it cannot distinguish one (Ch. H) – the same pattern-versus-occupation logic as in Doc. 293.

The caveat. The computation uses the smooth Laplacian, i.e. $D_f = 4$ exactly. All numbers are therefore *bare*; Ch. I separates what that costs: counts and ratios stay invariant, absolute energies carry $-1.33\,\%$, and whether the degeneracies split under the fractal perturbation has not been computed.

In one sentence: in Hilbert space the lattice becomes flat but not inconsequential – what geometry carried is carried by spectrum and fibre, and the division of labour between them is exactly determinable, as long as the smooth approximation is booked as such.

Notation

The table explains the symbols of this document. Quantities appearing only once are introduced there.

Note on notation. In lattice theory Λ is the usual symbol for a lattice (Conway/Sloane). In the FFGFT corpus, however, Λ is taken – as the cosmological constant, in ΛCDM, and in particular as the closure scale $\Lambda^* = 1/R_H^2$ (Doc. 312). This document therefore uses Γ throughout for the lattice and Γ^* for the dual lattice. Likewise L^* is *not* used here, since in Doc. 312 it denotes the cycle length; the torus edge lengths are called R_{123} and R_4.

Symbol	Meaning
Lattice and torus	
Γ	lattice in $\mathbb{R}^4$; defines the torus $\mathrm{T}^4 = \mathbb{R}^4/\Gamma$ (instead of the Λ customary in the lattice literature, see note on notation)
Γ^*	dual lattice; carries the admissible wave vectors of the Fourier modes
$D4$	root lattice D_4: all integer k with $\sum k_i$ even; kissing number 24
Z^4	cubic lattice $\mathbb{Z}^4$; kissing number 8, exactly self-dual
$D4^*$	dual D4 lattice; again a D4 lattice up to scaling
R_{123}, R_4	radii of the three spatial directions and of the fourth direction
r	radius ratio $r = R_4/R_{123}$ (dimensionless)
Spectrum	
k	wave vector (mode index), $k \in \Gamma^*$
k_4	fourth component of k; $k_4 = 0$ separates the FCC half from the winding half
$\lvert k\rvert^2$	norm; also the eigenvalue of the Laplacian on the isotropic torus
$E(k)$	eigenvalue under anisotropy, $E = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$
$N(n)$	degeneracy of the shell $\lvert k\rvert^2 = n$; coefficient of the theta series
θ_{D4}	theta series of the D4 lattice, $\theta_{D4} = (\theta_3^4 + \theta_4^4)/2$
$\theta(t)$	heat-kernel theta function, $\theta(t) = \sum_k e^{-t\lvert k\rvert^2}$
$\zeta_M(s)$	Epstein zeta function of the lattice M; $s = -1/2$ is the Casimir point

Symbol	Meaning
Symmetries	
Aut(Γ)	automorphism group of the lattice (length-preserving self-maps)
$W(F_4)$, $W(D4)$	Weyl groups; $\lvert W(F_4)\rvert = 1152$, $\lvert W(D4)\rvert = 192$
A	element of order 3 in Aut($D4$); half-integral, hence computed exactly via $2A$
det$(1-A)$	determinant; $\lvert$det$(1-A)\rvert = 9$ counts the orbifold fixed points
Z_3, C_3	cyclic group of order 3 (triality resp. fibre action)
S_3, A_5	symmetric group on 3 elements; alternating group on 5 elements (icosahedral)
ω	third root of unity, $\omega = e^{2\pi i/3}$; phase $+120^\circ$
χ_0, χ_1, χ_2	the three Z_3 characters; index the orbifold sectors
Hilbert space and fibre	
$\mathscr{H}$	state space, $\mathscr{H} = L^2(\mathrm{T}^4) \otimes \mathbb{C}^3$ (Docs. 230/282)
$\mathbb{C}^3$	fibre over each point; carries the Z_3 structure
3, $3'$	the two triplets of the doubling $6 = 3 \oplus 3'$ (Doc. 285)
φ	golden ratio, $\varphi = (1+\sqrt{5})/2$; occurs only in the A_5 layer
$\theta = 2/9$	phase share from Doc. 293; not a lattice invariant (Ch. F)
FFGFT references	
$\xi = 4/30000$	geometric fundamental constant of FFGFT
$\tilde{T} \cdot m = 1$	time-mass relation; readable in the spectrum as the splitting at $r \neq 1$
Status labels	
[K]	core derivation: computed and controlled
[S]	structural statement: consistent but not derived
theorem	mathematically proven, not merely checked numerically
boundary	explicitly booked limitation of scope

Chapter A

The starting position: flatness is not the difference

The torus $\mathsf{T}^4 = \mathbb{R}^4/\Gamma$ is *always* flat, whatever lattice Γ defines it. The D4 torus and the Z^4 torus both have zero curvature. The difference lies not in curvature but in lattice structure: kissing number 24 against 8, packing density $\pi^2/16$ against $\pi^2/32$.

On passing to $\mathscr{H} = L^2(\mathsf{T}^4)$ the lattice becomes the **index set of the Fourier modes**: the basis consists of plane waves $e^{ik\cdot x}$, and the admissible k lie on the dual lattice Γ^*. In this sense every lattice becomes flat in Hilbert space – it is no longer a sphere packing in space but a point lattice of quantum numbers.

The geometry is not lost; it migrates. The Laplacian has eigenvalues $|k|^2$, and the degeneracy of each eigenvalue is the number of lattice vectors of that length – the theta series of the lattice. The 24 shortest D4 vectors become the 24-fold degeneracy of the first excitation, and the decomposition $24 = 12 + 12$ of Doc. 311 (FCC neighbours plus winding neighbours) remains exactly readable as a k_4 split.

Two anchors keep this translation stable:

1. **D4 is self-dual up to scaling** – $D4^*$ is again a D4 lattice. The Fourier transform does not lead out of the family. (Z^4 is exactly self-dual.)
2. **The fibre changes nothing about the lattice.** In $\mathscr{H} = L^2(\mathsf{T}^4) \otimes \mathbb{C}^3$ the $\mathbb{C}^3$ fibre hangs at every point and carries the Z_3 structure; the mode lattice remains Γ^*.

$$\boxed{\text{geometry} \longrightarrow \text{spectrum} + \text{fibre}} \tag{A.1}$$

Chapter B

The spectrum separates what curvature does not [K]

Complete shell count up to $|k|^2 = 12$:

Norm $\|k\|^2$	Z^4	D4	D4 split (k_4=0 / k_4≠0)
1	8	0	—
2	24	24	12 / 12
3	32	0	—
4	24	24	6 / 18
5	48	0	—
6	96	96	24 / 72
7	64	0	—
8	24	24	12 / 12
9	104	0	—
10	144	144	24 / 120
11	96	0	—
12	96	96	8 / 88

Three readings:

(i) Two flat tori are spectrally distinguishable. Z^4 starts at norm 1 with degeneracy 8, D4 at norm 2 with degeneracy 24. For D4 the odd norms are *entirely* absent – the parity condition $\sum k_i$ even excludes them. What curvature does not distinguish, the spectrum distinguishes at once.

(ii) Independent control passed. The D4 series $24, 24, 96, 24, 144, 96 \ldots$ is the known theta series $\theta_{D4} = (\theta_3^4 + \theta_4^4)/2$; the script reproduces it coefficient by coefficient.

(iii) The $12 + 12$ decomposition is spectrally addressable. The first excitation carries exactly twelve modes with $k_4 = 0$ (the spatial FCC half) and twelve with $k_4 \neq 0$ (the winding half). On higher shells the split is asymmetric ($6/18$ at norm 4, $8/88$ at norm 12): **the $12/12$ symmetry is a property of the first shell, not a general theorem.** This must be booked explicitly – Doc. 311 argues with $24 = 12 + 12$, and that statement holds exactly where it is made.

Chapter C

Anisotropy: where the time-mass relation becomes readable [K]

If the fourth direction is compactified with a different radius ($r = R_4/R_{123}$, eigenvalue $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$), the first shell splits:

r	E (winding, 12 modes)	E (FCC, 12 modes)
1.0	2.000	2.000 (degenerate: 24)
1.1	1.826	2.000
1.5	1.444	2.000
2.0	1.250	2.000

The FCC modes stay fixed at $E = 2$, the winding modes follow exactly

$$E_{\text{winding}} = 1 + \frac{1}{r^2}, \qquad \Delta = 1 - \frac{1}{r^2}. \tag{C.1}$$

The splitting is a direct measure of the radius ratio – **the place where the $\tilde{T} \cdot m = 1$ structure becomes readable in the spectrum**: mass direction and spatial directions carry different excitation energies as soon as they have different lengths.

C.1 The crossing map is discrete and short

All levels have the form $E = a + b/r^2$ with integer a, b. In the range $1 < r \leq 2.5$, among all levels up to norm 8 there are **exactly two** crossing radii:

$r = \sqrt{2}$	pure winding $(0, 0, 0, \pm 2)$ crosses the FCC-12
$r = \sqrt{3}$	pure winding crosses the winding-12

The level sequence therefore does not change continuously and chaotically but at two distinguished root radii. Any ordering statement about the lowest excitations falls into exactly three regimes: $r < \sqrt{2}$, $\sqrt{2} < r < \sqrt{3}$, $r > \sqrt{3}$. Beyond $\sqrt{3}$ the *ground tone* of the spectrum is a pure winding (at $r = 2$: $E = 1.00$ with 2 modes before 1.25 with 12 before 2.00 with 12).

Chapter D

Reorganisations: three levels and one boundary

Flat lattices can be reorganised – on three levels that cost very different things.

Change of basis (free). Any unimodular transformation from $GL(4,\mathbb{Z})$ describes the same point lattice with different generators. The spectrum does not change at all. Pure bookkeeping.

Lattice symmetries (free, structure-bearing). Reorganisations mapping the lattice onto itself – this is where the central finding sits, Ch. E.

Rebuilding into another lattice (costs physics). Discretely via glue vectors along the chain

$$D4 \subset \mathrm{Z}^4 \subset D4^* \qquad \text{(index 2 each)}, \tag{D.1}$$

or continuously by deformation in the moduli space $GL(4,\mathbb{R})/(O(4)\times GL(4,\mathbb{Z}))$. Each step shifts degeneracies; D4 is a distinguished point there (kissing number 24, provably unbeatable in 4D).

The boundary of the spectral statement. From dimension 4 on there exist pairs of flat tori that are *isospectral but not isometric* (Schiemann). The general theorem "the spectrum determines the lattice" is *false* in 4D. Uncritical for D4 itself – kissing number 24 characterises it uniquely – but it must not be booked as a rule.

Chapter E

Triality is built in, not attached [K]

E.1 Computed on the lattice

$\lvert\mathrm{Aut}(D4)\rvert$	$1152 = \lvert W(F_4)\rvert$
$\lvert\mathrm{Aut}(\mathrm{Z}^4)\rvert$	$384 = 2^4 \cdot 4!$
$\lvert\mathrm{Aut}(D4)\rvert/\lvert W(D4)\rvert$	$1152/192 = 6 = \lvert S_3\rvert$

Both orders were determined by brute-force enumeration of all basis tuples with identical Gram matrix and match the known values. The quotient $6 = |S_3|$ establishes triality *directly on the lattice automorphism group*, not merely on the Dynkin diagram: D4 is the only lattice of the D_n family with three equivalent Dynkin arms; for $D5, D6, \ldots$ the outer symmetry shrinks to Z_2.

E.2 The 80 order-3 elements in three classes

Computed exactly (the triality elements are *half-integral* – the glue-vector mixture $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ enters; rounding to integers destroys them, hence the computation uses $2A$):

Class	Count	Character
trace -2, half-integral	16	fixed-point free in $\mathbb{R}^4$; eigenvalues $\{\omega, \bar{\omega}\}$ twice
trace 1, integral	32	fixed plane in $\mathbb{R}^4$, free on the 24 minimal vectors
trace 1, half-integral	32	6 fixed vectors + 6 triple orbits

E.3 Two consequences

(a) The orbifold is a lattice symmetry, not an extra assumption. For each of the 16 elements with trace -2 one has exactly

$$|\det(1 - A)| = 9, \tag{E.1}$$

so the quotient $\mathrm{T}^4/\langle A\rangle$ is a Z_3 orbifold with *exactly 9 fixed points* – the standard number 3^2. All nine lie at third-coordinates. **The $\mathbf{T}^4/Z_3$ structure need not be imposed; it is already present in D4.**

(b) On every mode orbit the Z_3 acts as a circulant. The unitary action on $L^2(\mathrm{T}^4)$ permutes the three plane waves of a triple orbit cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases 0°, $+120^\circ$, -120° – *structurally exactly the circulant diagonalisation of the $\mathbb{C}^3$ fibre* (Doc. 291). The first shell provides eight copies: $24 = 8 \times 3 = 8 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$.

Chapter F

The negative finding: 5 does not divide 1152 [K]

The order distribution in Aut($D4$), determined exactly:

Order	1	2	3	4	6	8	12
Count	1	139	80	228	464	144	96

Total 1152. **Order 5: zero elements** – and that is not a computational result but a theorem: $1152 = 2^7 \cdot 3^2$, and 5 does not divide it (Lagrange).

Consequence for $\theta = 2/9$. The fivefold rotation R_5, from which $p_0 = |\langle v_0|R_5 v_e\rangle|^2 = 2/9$ follows (Doc. 293), is *in principle* incompatible with the D4 lattice structure. The test was additionally carried out directly: all invariants of the linear lattice action at the nine orbifold fixed points ($x{\cdot}x$, $k{\cdot}x$, $x{\cdot}y$, all mod 1 in exact rational arithmetic) have denominator spectrum $\{1, 2, 3, 6\}$ – **no ninths**. Notably, $x{\cdot}y$ is a multiple of one ninth, but the numerators are always divisible by 3; the D4 geometry *suppresses* ninths at the first level.

What this means – and what it does not. $2/9$ is not an invariant of the lattice action. This does not devalue Doc. 293 but delimits it: the phase belongs to a layer *above* the lattice (the A_5 structure), the lattice carries the Z_3. The division of labour is thereby exactly named rather than surmised.

Chapter G

The doubling is nevertheless prepared [K]

Does the negative finding contradict the doubling $6 = 3 \oplus 3'$ of Doc. 285? No – in three specifiable stages.

(1) The prohibition hits the wrong level. $5 \nmid 1152$ forbids order-5 *lattice automorphisms*. The doubling lives on the **fibre** ($\mathbb{C}^3 \to \mathbb{C}^6$); A_5 is nowhere required as a spatial symmetry. The compatibility condition is only that the interface of the two levels exists – and that is exactly determinable: element orders in A_5 are $\{1, 2, 3, 5\}$, in Aut$(D4)$ they are $\{1, 2, 3, 4, 6, 8, 12\}$; the intersection is $\{1, 2, 3\}$, canonically C_3. **And exactly that is provided by triality.** The interface is not merely permitted, it is the only possible one – and it is occupied.

(2) The lattice prepares the doubling. The central involution $-1 \in$ Aut$(D4)$ commutes with the triality Z_3 and pairs the eight triple orbits of the first shell into **four disjoint pairs**. No orbit is self-paired, provably: $-k = A^j k$ would require eigenvalue -1 for an element of order 3 – impossible. Hence

$$24 = 8 \times 3 = 4 \times 6 \qquad \text{(orbit } \oplus \text{ mirror orbit).} \tag{G.1}$$

Each six-block carries under C_3 the character content $2\times(\chi_0 \oplus \chi_1 \oplus \chi_2)$ – exactly the restriction of $3 \oplus 3'$. The first excitation shell provides four ready containers.

(3) The boundary, cleanly booked. The *labelling* – which factor is 3 and which $3'$ – is lattice-blind. The A_5 characters of the two triplets differ only on the 5-classes (φ against $1 - \varphi$); on the 3-class both are zero and the C_3 restrictions are identical. The distinction hangs exactly on the structure that does not exist in the lattice – fully consistent with Ch. F.

Balance: the lattice provides containers and C_3 content, the A_5 layer provides the $3/3'$ labels, φ and the fivefold – and hence $2/9$.

Chapter H

Anchor-free results [K]

All quantities below are dimensionless – ratios, counts, congruences. No Kammerton, no SI scale enters.

H.1 Two congruence theorems for the D4 theta series

All complete shells up to norm 120 checked:

$$N(n) \equiv 0 \pmod 3 \qquad \text{and} \qquad N(n) \equiv 0 \pmod 6 \qquad \text{without exception.} \tag{H.1}$$

Both are theorems, not empirics: the triality Z_3 is fixed-point free on $\Gamma \setminus \{0\}$ (eigenvalues $\omega, \bar{\omega}$ – no eigenvalue 1), so *every* shell decomposes into triple orbits; together with the -1 pairing without self-pairing, into six-blocks.

H.2 Character-resolved T^4/Z_3 spectrum

Consequence: exact equidistribution on every shell.

Norm	$N(D4)$	χ_0	χ_1	χ_2
2	24	8	8	8
4	24	8	8	8
6	96	32	32	32
10	144	48	48	48
14	192	64	64	64
18	312	104	104	104

The invariant sector of the orbifold (χ_0) has exactly $N(n)/3$ states per shell; the Z_3 projection costs no asymmetry on any shell.

Honest consequence. Equidistribution and six-containers are *not* a distinction of the first shell but a theorem for all. **The structure is everywhere – and therefore cannot select anything.** This is exactly the pattern-versus-occupation logic of Doc. 293: the pattern is present scale-invariantly; which shell is physically occupied is decided by something else.

H.3 Casimir: the densest lattice loses

ζ-regularised vacuum energy of a scalar field, $E = \pi\zeta_M(-1/2)$, both tori normalised to covolume 1 (Epstein zeta via the Riemann split representation, verified three times – against direct summation at $s = 4, 5$ and against the closed form $Z_{\mathbb{Z}^4}(s) = 8(1-4^{1-s})\zeta(s)\zeta(s-1)$, agreement to 25 digits):

	$\zeta_M(-1/2)$	$E = \pi\zeta_M(-1/2)$
Z^4 torus	-0.2966893	-0.9320768
D4 torus	-0.2764778	-0.8685805
Difference		$+0.0634963$

The Z^4 torus lies lower. The reason is the spectral gap: at equal volume the densest lattice D4 has the *largest* first eigenfrequency ($\sqrt{2}$ against 1) and hence the *less* negative zero-point energy. Notable is the reversal of ordering: for $s > 0$ D4 minimises the Epstein zeta (the known extremality of the densest lattice), at $s = 0$ all lattices agree ($\zeta(0) = -1$ universally), and at $s = -1/2$ the ordering is inverted.

Interpretation boundary, booked: this holds for the scalar, periodic field. Different field content or different boundary conditions can change the sign of the preference – "nature chooses Z^4" does *not* follow.

H.4 Heat kernel: the observability limit quantified

For both tori $t^2\theta(t) \to 1$ (identical Weyl term at equal volume); the lattice difference is exponentially small:

t	relative difference
0.5	1.2×10^{-2}
0.2	1.2×10^{-6}
0.1	1.8×10^{-13}
0.05	4.1×10^{-27}

Any observable resting only on the leading heat-kernel coefficients is lattice-blind. The D4/Z^4 difference sits exclusively in the exponentially suppressed theta-series fine structure – or directly in the spectral gap.

Chapter I

What is still bare here [K]/open

This document computes with the *smooth* Laplacian on $\mathrm{T}^4 = \mathbb{R}^4/\Gamma$, i.e. with $D_f = 4$ exactly. FFGFT, however, carries the fractal dimension $D_f = 4 - \xi$ and the factor $K_{\text{frak}} = 1 - 100\xi$ (A040). All numbers here are therefore **bare** – in the same sense in which E_0 was bare before the clarification in register R72. The consequence falls into three very unequal parts.

I.1 Untouched: everything anchor-free

By the assignment rule (register R70, Doc. 313), K_{frak} applies to absolute quantities and cancels for ratios of the same level. Untouched therefore:

- all **counts** – 24, 8, $N(n)$, 8×3, 4×6: integer combinatorics, no length;
- the **congruences** $N(n) \equiv 0$ mod 3 and mod 6: group theory, no metric content;
- the **character split** $N/3$ per sector;
- all **ratios** E/E_{min} and the crossing radii $\sqrt{2}$, $\sqrt{3}$ – structure over structure, the factor cancels.

This is the real reason why Ch. H is called "anchor-free results": those statements are fractal-invariant.

I.2 One factor: all absolute energies

With an anchor, $E = \hbar H_0\,|k|$ (ground mode $= \hbar H_0$, see Ch. B together with Doc. 312). These values are bare and carry the factor:

	bare	with K_{frak}
first D4 shell ($\|k\| = \sqrt{2}$)	3.230×10^{-52} J	3.187×10^{-52} J
pure winding ($\|k\| = 2$)	4.567×10^{-52} J	4.507×10^{-52} J

Throughout $-1.33\,\%$. The same holds for the Casimir energies of Ch. H and for the spectral gap itself. Bookkeeping, settled; not structurally interesting.

I.3 Open: the operator, not the conversion

The third part cannot be settled by multiplication. At $D_f = 4 - \xi$ the operator is *no longer a smooth Laplacian* – and exact symmetry is precisely what carries the degeneracies. A fractal perturbation breaks it:

Perturbation strength	$\Delta E/E$	ΔE at $E = \hbar H_0$
$O(\xi)$	1.3×10^{-4}	3.0×10^{-56} J
$O(100\xi) = 1 - K_{\text{frak}}$	1.3×10^{-2}	3.0×10^{-54} J

The 24-fold degeneracy would then no longer be a degeneracy but a multiplet with fine structure. Which sub-multiplets it decomposes into is decided by the *symmetry of the perturbation*:

Perturbation respects	Splitting of the first shell
triality (Z_3) and -1	24 stays together
only Z_3	8 triple orbits stay together
only -1	4 six-blocks (Ch. G)
neither	up to 24 individual levels

This has not been computed. It would be the natural continuation of this document and no large calculation – first-order perturbation theory on a finite matrix, no Boltzmann apparatus. Until then: the spectrum of this document is the *smooth approximation*, and the container structure is only as stable under perturbation as the symmetry that carries it.

Chapter J

Status overview

Statement	Content	Status
Translation	geometry → spectrum + fibre; lattice becomes mode index, degeneracies = theta series	[K]
Distinguishability	Z^4 from norm 1 (8), D4 from norm 2 (24); odd norms empty for D4	[K]
$12 + 12$	first shell exactly FCC + winding; higher shells asymmetric ($6/18$, $8/88$)	[K]
Anisotropy	$E_{\text{wind}} = 1 + 1/r^2$; crossings only at $r = \sqrt{2}$ and $\sqrt{3}$; three regimes	[K]
Triality	$\lvert\text{Aut}(D4)\rvert = 1152$, quotient $6 = \lvert S_3\rvert$; 80 order-3 elements in three classes	[K]
Orbifold	16 elements with $\lvert\det(1 - A)\rvert = 9$: T^4/Z_3 is a lattice symmetry, not an extra assumption	[K]
Circulant	Z_3 acts on every triple orbit with $\{1, \omega, \omega^2\} = \mathbb{C}^3$ fibre structure	[K]
$5 \nmid 1152$	no order-5 automorphisms; $2/9$ is not a lattice invariant (denominators $\{1, 2, 3, 6\}$)	[K] theorem
Doubling	interface $A_5 \cap \text{Aut}(D4) = C_3$, occupied; $24 = 4 \times 6$ as containers	[K]
Labelling	3 versus $3'$ is lattice-blind (difference only on 5-classes)	[K] boundary
Congruences	$N(n) \equiv 0 \bmod 3$ and $\bmod 6$ on all 60 shells checked, with proof ground	[K] theorem
Selection	containers and equidistribution are generic – the structure cannot distinguish any shell	[K]
Casimir	$E(Z^4) = -0.9321$ against $E(D4) = -0.8686$; ordering reversed relative to $s > 0$	[K], field-content dependent
Heat kernel	Weyl term identical, difference 4×10^{-27} at $t = 0.05$: leading order is lattice-blind	[K]
Spectral theorem	“spectrum determines lattice” false in 4D (Schiemann); uncritical for D4, but not a rule	[K] boundary
Fractal status	counts/ratios invariant; absolute energies $-1.33\,\%$; degeneracy splitting uncomputed	[K]/open

Balance: in Hilbert space the lattice becomes flat but not inconsequential. The kissing number becomes the degeneracy, triality becomes the orbifold and the fibre, the radius ratio becomes the level splitting. What the lattice does *not* carry – fivefold symmetry and hence φ and $2/9$ – is determined just as exactly as what it does carry. And the container structure is generic: it provides, it does not select.

Chapter K

Check scripts

Three scripts, all deterministic, all controls passed:

- `d4_skript_1_spektrum_deformation.py` – theta series D4/Z^4, k_4 split, deformation r, exact crossing radii (standard library only).
- `d4_skript_2_trialitaet_orbifold_phasen.py` – Aut$(D4) = 1152$, order-3 classification (computed exactly with $2A$), $|\det(1-A)| = 9$, circulant, phase test $2/9$, order distribution, $24 = 4 \times 6$ (numpy).
- `d4_skript_3_schalen_casimir.py` – shell theorems, Epstein zeta at $s = -1/2$, heat kernel (numpy, mpmath).

Controls against known values: θ_{D4}, $|W(F_4)| = 1152$, $2^4 \cdot 4! = 384$, $|W(D4)| = 192$, orbifold fixed-point count $3^2 = 9$, Jacobi identity for $Z_{\mathbb{Z}^4}$.

Registration: entry into Doc. 190 will be done separately, should the delimitation against Doc. 293 be booked there.